$$= \frac{0.52 \times 6.5 \times 1000}{31.75 \times 100} = 1.06$$

$T_b - 100 = 1.06$

$T_b = 100 + 1.06$

$= 101.06 \approx 101^o$ C

13. $\frac{\Delta P}{P^o} = x_2$ (mole fraction of the solute)

option (b)

14. option (d) 0.1×3 ion $[Ba^{2+}, 2NO_3^-]$

0.1×3 ion $[2\ Na^+, SO_4^-]$

15. $(\pi_1)_{\text{non electrolyte}} = (\pi_2)_{\text{glucose}}$

$C_1RT = C_2RT$

$\frac{W_1}{M_1} = \frac{W_2}{M_2}$ $\qquad CH_2O$

$12 + 2 + 16 = 30$

$\frac{6}{n(30)} = 0.025$

$\therefore n = \frac{6}{0.025 \times 30} = 8$

$\therefore$ molecular formula $= C_8H_{16}O_8$ (option b)

16. $K_H = 4 \times 10^4$ atm

$\left(P_{O_2}\right)_{air} = 0.4$ atm

$\left(x_{O_2}\right)_{\text{in solution}} = ?$

$\left(P_{O_2}\right)_{air} = K_H \left(x_{O_2}\right)_{\text{in solution}}$

$0.4 = 4 \times 10^4 \left(x_{O_2}\right)_{\text{in solution}}$

$\Rightarrow \left(x_{O_2}\right)_{\text{in solution}} = \frac{0.4}{4 \times 10^4} = 1 \times 10^{-5}$ (Option c)

17. Normality of H_2SO_4 = (no. of replacable H^+) $\times$ M

$= 2 \times 1.25$

$= 2.5$ N

18. ΔH_{mix} is negative and show negative deviation from Raoults law.

option (d)

19. $\frac{\Delta P}{P^o} = x_{sugar}$

$3.5 \times 10^{-3} = x_{sugar}$

$x_{Sugar} + x_{H_2O} = 1$

$\therefore x_{H_2O} = 1 - 0.0035 = 0.9965$ (Option d)

20. $\frac{\Delta P}{P^o} = x_2$

$\frac{100 - 90}{100} = \frac{n_2}{n_2 + n_1}$

$\Rightarrow \frac{n_2 + n_1}{n_2} = \frac{100}{10}$ $\qquad \left[n_1 = \frac{92}{92} = 1\right]$

$1 + \frac{1}{n_2} = 10$

$\frac{1}{n_2} = 9$

$\Rightarrow n_2 = \frac{1}{9}$

$\Rightarrow \frac{W_2}{M_2} = \frac{1}{9}$

$\Rightarrow W_2 = \frac{M_2}{9}$

$W_2 = \frac{80}{9} = 8.89$ g

21. $\pi = CRT$

$y = x\ (m)$

$m = RT$

$310\ R = RT$

$\therefore T = 310$ K

$= 37^o$ C

option (c)

22. $\pi = CRT$

$\pi = \frac{W}{M \times V} \times RT$

$\therefore M = \frac{WRT}{\pi V}$

$= \frac{1.26 \times 0.083 \times 300}{2.52 \times 10^{-3} \times 0.2}$

$= 62.22$ Kg mol^{-1}

Key answers -jagan.indd 301 22-01-2021 13:23:32